ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

MATHEMATICS P1

MARKING GUIDELINE

MARKS: 100

This marking guideline consists of 6 pages.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept any correct alternative method. Give follow-through marks (CA) where a learner uses their own earlier wrong answer correctly. Answers only are not awarded full marks where working is required.

QUESTION 1

1.1.1 $= 2x^3 + 8x^2 - 2x - 3x^2 - 12x + 3$ ✓✓ (3)
$= 2x^3 + 5x^2 - 14x + 3$ ✓

1.1.2 $= \frac{(x-3)(x+3)}{(x+3)(x-2)} \times \frac{x-2}{x-3}$ ✓✓✓ (factorising, multiplying by the reciprocal) (4)
$= 1$ ✓

1.2.1 $= 3(x^2 - 4)$ ✓ $= 3(x-2)(x+2)$ ✓ (2)

1.2.2 $= (2x+3)(x-5)$ ✓✓ (2)

1.2.3 $= (x-2)$ ✓$(x^2 + 2x + 4)$ ✓ (2)

1.2.4 $= a(x-y) + 2(x-y)$ ✓ $= (x-y)(a+2)$ ✓ (2)

1.3.1 $= \frac{2^x(2^2-1)}{3 \cdot 2^x}$ ✓ $= \frac{2^x \cdot 3}{3 \cdot 2^x}$ ✓ $= 1$ ✓ (3)

1.3.2 $= (3^3)^{2/3} \times (2^2)^{-1/2}$ ✓ $= 9 \times \frac{1}{2}$ ✓ $= \frac{9}{2}$ (or 4,5) ✓ (3)

1.3.3 $= x^6y^{-4} \times x^{-4}y^5$ ✓ $= x^{6-4}y^{-4+5}$ ✓ $= x^2y$ ✓ (3)

[24]

QUESTION 2

2.1.1 $3x - 6 = 2x + 5$ ✓ $\therefore x = 11$ ✓ (2)

2.1.2 $(x-7)(x+2) = 0$ ✓ $\therefore x = 7$ ✓ or $x = -2$ ✓ (3)

2.1.3 $3(x+1) - 2(x-3) = 12$ ✓ (multiply by LCD 6) (3)
$3x + 3 - 2x + 6 = 12$ ✓ $\therefore x = 3$ ✓

2.1.4 $3^{x+1} = 3^4$ ✓ $\therefore x + 1 = 4, x = 3$ ✓ (2)

2.1.5 $x^2 - 3x - 10 = 0$ ✓ $(x-5)(x+2) = 0 \therefore x = 5$ ✓ or $x = -2$ ✓ (3)

2.2 $-2 < 2x \leq 8$ ✓ $\therefore -1 < x \leq 4$ ✓ (3)
Number line: open circle at −1, closed (shaded) circle at 4, line between them ✓

2.3 From $3x - y = 7$: $y = 3x - 7$ ✓ (4)
$x + 2(3x - 7) = 7$ ✓ $\rightarrow 7x = 21 \therefore x = 3$ ✓; $y = 3(3) - 7 = 2$ ✓

2.4 $A - P = Pin \checkmark \therefore i = \frac{A - P}{Pn} \checkmark$ (2)

[22]

QUESTION 3

3.1 $21 \checkmark ; 25 \checkmark$ (2)

3.2 $\text{Common difference } d = 4 \checkmark; T_n = 4n + 1 \checkmark$ (2)

3.3 $T_{50} = 4(50) + 1 \checkmark = 201 \checkmark$ (2)

3.4 $4n + 1 = 157 \checkmark \therefore n = 39, \text{it is the 39th term} \checkmark$ (2)

3.5 $4n + 1 = 100 \rightarrow n = 24{,}75 \checkmark$ (2)

n must be a natural number, so 100 is not a term ✓

[10]

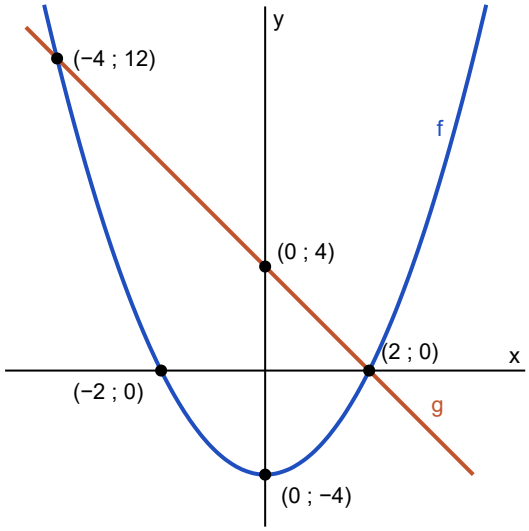
QUESTION 4

4.1 $(0 ; -4) \checkmark$ (1)

4.2 $x^2 - 4 = 0 \rightarrow x = \pm 2 \checkmark; (-2 ; 0) \text{ and } (2 ; 0) \checkmark$ (2)

4.3 $-2x + 4 = 0 \rightarrow (2 ; 0) \checkmark$ (1)

4.4 (5)



Marks: shape of *f* ✓; turning point (0 ; −4) ✓; *x*-intercepts of *f* ✓; straight line *g* with negative gradient ✓; intercepts of *g*: (0 ; 4) and (2 ; 0) ✓

4.5 $y \geq -4 \text{ (or } y \in [-4 ; \infty)) \checkmark$ (1)

4.6 $x^2 - 4 = -2x + 4 \rightarrow x^2 + 2x - 8 = 0$ ✓ (4)
$(x + 4)(x - 2) = 0 \rightarrow x = -4$ or $x = 2$ ✓
$f(x) \geq g(x)$ when $x \leq -4$ ✓ or $x \geq 2$ ✓

4.7 $q = 1$ ✓ (3)
$4 = \frac{a}{2} + 1$ ✓ $\therefore a = 6$ ✓

4.8.1 $y = 0$ ✓ (1)

4.8.2 $(0\ ;\ 1)$ ✓ (1)

[19]

QUESTION 5

5.1 $A = P(1 + in) = 8\ 000(1 + 0{,}07 \times 5)$ ✓✓ $= R10\ 800$ ✓ (3)

5.2 $A = P(1 + i)^n = 8\ 000(1{,}065)^5$ ✓✓ $= R10\ 960{,}69$ ✓ (3)

5.3 Sipho ✓, by R160,69 ✓ (2)

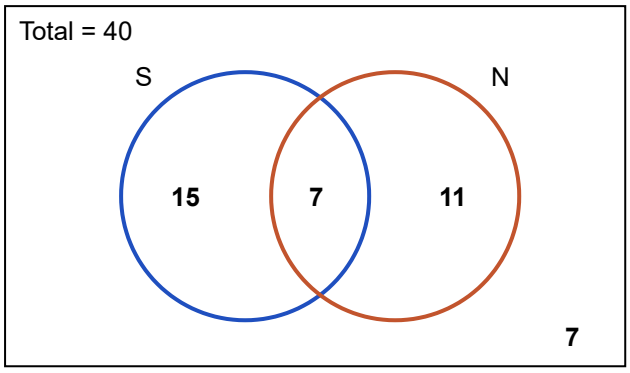
5.4 $A = 6\ 500(1{,}05)^3$ ✓✓ $= R7\ 524{,}56$ ✓ (3)

5.5 $\frac{2\ 300}{18{,}40}$ ✓ $= \$125$ ✓ (2)

[13]

QUESTION 6

6.1 (4)



Marks: both circles inside the rectangle ✓; 7 in the overlap ✓; 15 (only S) and 11 (only N) ✓; 7 outside both circles ✓

6.2.1 $P(\text{neither}) = \frac{7}{40}$ ✓✓ (2)

6.2.2 $P(\text{S or N}) = \frac{15+7+11}{40}$ ✓ $= \frac{33}{40}$ ✓ (2)

6.2.3 $P(\text{only S}) = \frac{15}{40}$ ✓ $= \frac{3}{8}$ ✓ (2)

6.3 No ✓. 7 learners play both, so S and N can happen together (n(S and N) ≠ 0) ✓ (2)

[12]

TOTAL: 100

END OF MARKING GUIDELINE